SCIENCE 1.4 **Name:** ***ANSWERS***

WORKSHEET THREE **ACIDS AND ALKALIS**

1. The taste of an acid is ***sour*** The taste of a base is ***bitter***
2. When red litmus is put into acids what happens? ***Stays the same***

   When blue litmus is put into acids what happens? ***Turns from blue to red***

   When red litmus is put into bases what happens? ***Turns from red to blue***

   When blue litmus is put into bases what happens? ***Stays the same***
3. Litmus is an example of an acid-base…. ***indicator***
4. The name of a vegetable that contains an indicator is … ***red cabbage***
5. The name of a flower that contains an indicator is… ***camellia, rose,***
6. An indicator is a substance that changes ***colour*** in acids and bases.
7. In what major way is Universal indicator different from Litmus?

   Litmus has only two colours, red and blue. UI has a range of colours
8. What colour is U.I. in pure water. ***green***
9. In order to describe how acidic an acid is, or how basic a base is we use the ***pH*** scale.
10. Universal indicator turns different colours depending on the ***pH*** of the solution.
11. Water has a pH of ***7*** and is said to be… ***neutral***
12. If a chemical has a pH of 3 it would be an …***acid***
13. If a chemical had a pH of 13 it would be an … ***alkali***
14. Chemical A has a pH of 9 and chemical B has a pH of 12. A and B are both ***bases***

    (acids/bases) but B is a ***stronger*** (stronger / weaker) ***base*** (acid / base) than A.
15. Write the names and formulae of the three **strong mineral acids** we use in the laboratory. (M)

    hydrochloric acid HCl ***sulfuric acid H_2SO_4***

    nitric acid HNO_3
16. Write the name and formula of the **weak organic acid** present in vinegar.

    Ethanoic acid CH_3COOH
17. What is the definition of a base? ***A base is a substance that neutralises an acid***

18. What is the definition of an alkali? ***A soluble base***

19. (a) When you put a zinc into acid an invisible gas called ***hydrogen*** is made.

(b) This gas is very ***light*** (light / heavy).

(c) To test for this gas we put a burning ***splint*** into it and there is a ***popping*** sound.

20. What is the name and formula of a common strong alkali used in the lab.

Sodium hydroxide NaOH

21. We often refer to sodium hydroxide as **caustic soda**. What is the meaning of **caustic**?

Capable of burning, dissolving or corroding by chemical action.

22. A common weak alkali used in the lab is actually a solution made by dissolving a smelly gas in water. What is it called? ***Ammonia***

23. Two solutions have lost their labels but one of them is vinegar and the other is caustic soda. Explain how you would discover which was which? ***Add an acid-base indicator to each solution. Vinegar will be the acid and caustic soda will be the base.***

24. Some text books suggest that litmus indicator has a **purple** colour in water. How would you account for this? ***The purple colour is a mixture of the red and blue colours which are present in equal amounts in neutral water***.

25. A solution of hydrochloric acid has a pH of 1. A large volume of water is added to the acid and mixed thoroughly. Suggest if there will be any change in the pH of the mixture. ***As the acid is diluted its pH will rise*** Use this idea to suggest how acid spills should be treated. ***Add lots of water*** Suggest another chemical that could be added to reduce the danger of an acid spill. ***Sodium bicarbonate / sodium carbonate/ a weak base***

26. Bases are sometimes classified as the oxides and hydroxides of metals. Use this idea and what you know about acids to help you circle the substances in the list that are bases.

NaCl (***NaOH***) (***K_2O***) (***$Zn(OH)_2$***) ***HCl*** ***HNO_3*** (***CuO***) ***MgSO4*** (***MgO***) (***$Fe(OH)_2$***)

27. Circle the substances in the following list that are acids.

Li_2SO_4 ***$AlCl_3$*** (***HNO_3***) ***NH_3*** ***Fe_2O_3*** (***H_2SO_4***) (***CH_3COOH***) ***NaOH*** (***HCl***) (***K_2O***)

28. When acids are reacted with carbonates and bicarbonates a colourless gas is given off.

(a) What is the name of this gas? ***Carbon dioxide***

(b) Describe how you would test for this gas. ***Turns limewater milky***